Chance B. Ronemus

chanceronemus.com chance.ronemus@ucalgary.ca +1 (805) 610-4630
0000-0001-6345-9589 Google Scholar

Research Interests

Orogenic Systems and Cordilleran Tectonics; Forearc and Foreland Basin Evolution; Climate-Tectonics Interactions and Stable Isotope Paleoaltimetry; Fluvial, Alluvial, and Deep-Water Depositional Systems; Zircon Geo-, Petro-, and Thermochronology

Education

University of Arizona — Tucson, AZ, USA
Ph.D., Geoscience (Advisor: Peter G. DeCelles) — 2025
Dissertation: Cretaceous to Cenozoic Evolution of the Southern Central Andes and Basins Therein

Montana State University — Bozeman, MT, USA
M.S., Geology (Advisor: Devon A. Orme) — 2021
Thesis: Orogens of Big Sky Country: Reconstructing the deep-time tectonothermal histories of the Beartooth and Highland mountains, southwestern Montana

Montana State University — Bozeman, MT, USA
B.S., Geology (Highest Honors) — 2019

Appointments

Assistant Professor (Tenure-Track) — Calgary, AB, Canada
University of Calgary, Dept. of Earth, Energy & Environment — 2026–present

Visiting Postdoctoral Scholar — Stanford, CA, USA
Stanford University (Mentor: Stephan A. Graham) — Summer 2026

NSF Earth Science Postdoctoral Fellow — Pocatello, ID, USA
Idaho State University (ISU; Mentor: Kurt E. Sundell) — 2025–2026

Fulbright Research Scholar — Mendoza, Argentina
IANIGLA-CONICET CCT — 2024

Publications

Peer-reviewed Publications [6 articles (3 first-author), 1 geologic map]

First-Author Journal Articles

7. **Ronemus, C.B.**, C.J. Howlett, P.G. DeCelles, B. Carrapa, and S.W.M. George (2024). The Manantiales Basin, southern Central Andes (~32 °S), preserves a record of Late Eocene–Miocene episodic growth of an east-vergent orogenic wedge. *Tectonics* 43.3. DOI: 10.1029/2023TC008100.

6. **Ronemus**, **C.B.**, D.A. Orme, W.R. Guenthner, S.E. Cox, and C.A.L. Kussmaul (2023). Orogens of Big Sky country: Reconstructing the deep-time tectonothermal history of the Beartooth Mountains, Montana and Wyoming, U.S.A. *Tectonics* 42.1. DOI: 10.1029/2022TC007541.

5. **Ronemus**, **C.B.**, D.A. Orme, S. Campbell, S.R. Black, and J. Cook (2021). Mesoproterozoic–Early Cretaceous provenance and paleogeographic evolution of the northern Rocky Mountains: Insights from the detrital zircon record of the Bridger Range, Montana, U.S.A. *GSA Bulletin* 133.3–4, 777–801. DOI: 10.1130/B35628.1.

Contributing Author Journal Articles

4. George, S.W.M., B. Carrapa, P.G. DeCelles, G. Jepson, H. Nadoya, C. Tabor, C.J. Howlett, **C.B. Ronemus**, M.T. Clementz, and L. Schoenbohm (2025). Increased moisture availability in the Central Andes during the Miocene Climatic Optimum. *Palaeogeography, Palaeoclimatology, Palaeoecology* 663, 112732. DOI: 10.1016/j.palaeo.2025.112732.

3. Howlett, C.J., **C.B. Ronemus**, B. Carrapa, and P.G. DeCelles (2025). Miocene construction of the High Andes recorded by exhumation of the Frontal Cordillera, La Ramada Massif of western Argentina (32 °S). *Tectonics* 44.1, e2024TC008433. DOI: 10.1029/2024TC008433.

2. Romero, M.C., D.A. Orme, K.D. Surpless, **C.B. Ronemus**, and Z. Morrow (2024). Age and provenance relationships between the basal Great Valley Group and its underlying basement: Implications for initiation of the Great Valley forearc basin, California, U.S.A. *Journal of Sedimentary Research* 94.5, 641–662. DOI: 10.2110/jsr.2024.004.

Geologic Map Sheet

1. **Ronemus**, **C.B.** and D.A. Orme (2023). *Geologic map of the eastern half of the Melrose 7.5' quadrangle and the western half of the Wickiup Creek 7.5' quadrangle, southwestern Montana*. 1 sheet, scale 1:24,000, EDMAP 16. Butte, MT: Montana Bureau of Mines and Geology.

Book Chapter (Popular Press)

1. **Ronemus**, **C.B.** (2025). "Regional geology: A journey through deep time". In: Kussmaul, C. *Backcountry Skiing: Peaks and Couloirs of Southwest Montana*. 2nd ed. (1st ed. 2020). PKI Press, pp. 14–18. ISBN: 978-1-7364316-2-7.

Submitted Manuscripts

4. **Ronemus**, **C.B.**, C.J. Howlett, P.G. DeCelles, B. Carrapa, A. Echaurren, M. Barrionuevo, J.G. Mosolf, M.L. Foley, and M.N. Ducea. Building the Andean Crust at ∼35 °S: Arc Geochemistry Reveals Cretaceous Thinning and Rapid Neogene Thickening. In Review with *Journal of Geophysical Research: Solid Earth*.

3. **Ronemus**, **C.B.**, C.J. Howlett, P.G. DeCelles, B. Carrapa, L.M. Fennell, K. Thirumalai, V.A.P. Muller, and L. Lothari. Miocene surface uplift and moisture-source evolution resolved by stable isotope proxies in the southern Central Andean foreland basin (∼32°S). In Review with *Earth and Planetary Science Letters*, Spring 2026.

2. Muller, V.A.P., B. Carrapa, S. Thomson, P.G. DeCelles, **C.B. Ronemus**, C.J. Howlett, and S.L. Beck. Late Miocene headwater erosion of the Central Andes drives fold-and-thrust belt exhumation. In Review with *Earth and Planetary Science Letters*, Spring 2026.

1. Orme, D.A., C.M. Baird, **C.B. Ronemus**, M.A. Malkowski, A.K. Laskowski, and F. Foster. Investigation of the sedimentary processes, depositional environments, and provenance of the Mesoproterozoic Belt Supergroup rocks in the Whitehall area, southwest Montana. In Review as a book chapter in a *GSA Books* volume honoring D. Winston.

Invited Presentations

- Building the High Andes: The view from the foreland basin record, *Idaho State University* 2025
- Cordilleran evolution in the sedimentary basin record, *University of Alaska Fairbanks* 2025
- Mapping as a foundation for understanding orogenic systems, *MT Bureau of Mines and Geology* .. 2025
- Tectonic controls on Cu mineralization in the Central Andes, *Teck Resources Structural Geology Team* 2025
- Cenozoic evolution of the southern Central Andean foreland basin, *University of Buenos Aires* ... 2024

Grants & Awards (21 awards; $354,295 total research funding)

Major Fellowships

- Earth Science Postdoctoral Research Fellowship ($291,976), *US National Science Foundation* 2025
- Fulbright Research Scholarship ($10,700 + other support), *US Fulbright Program* 2024

Research Grants and Other Awards

- Undergraduate Student Research Funding Award (PI of record: K. Sundell; $6,000), *Idaho Higher Education Research Council* 2025
- John and Nancy Sumner Scholarship ($1,000), *UArizona Dept. of Geosciences* 2025
- Lewis & Clark Exploration and Field Research Scholarship ($4,900), *Amer. Philosophical Society* . 2024
- Research and Project Grant ($1,490), *UArizona Graduate and Professional Student Council* 2023
- Travel Grant ($350), *GSA Cordilleran Section* 2023
- Coney Scholarship ($3,000), *UArizona Dept. of Geosciences* 2023
- Galileo Circle Scholarship ($1,000 each), *UArizona College of Science* 2022, 2023
- Travel Grant ($1,000), *UArizona Graduate and Professional Student Council* 2022
- 1st place, Best Student Geologic Map Competition, *US Geological Survey* 2021
- Sulzer Scholarship ($4,676), *UArizona Dept. of Geosciences* 2021
- StraboSpot Super Tester Award ($1,500), *University of Kansas* 2021
- Open Access Author Fund Award ($2,000), *MSU Library* 2020
- Harrison Scholarship ($500), *Tobacco Roots Geological Society* 2020
- Donald L. Smith Memorial Scholarship ($3,000), *MSU Dept. of Earth Sciences* 2020
- EDMAP Grant (PI of record: D. Orme; $17,428), *US Geological Survey* 2020
- Research Grant ($750), *Montana Academy of Sciences* 2019
- Research Grant ($225), *Tobacco Roots Geological Society* 2019
- Undergraduate Scholars Program Research Grant ($1,800), *MSU* 2018

Teaching

Instructor of Record

- Solid Earth, *ISU* .... Spring 2025
- Independent Problems and Studies in Geology, *ISU* .... Spring 2025

Teaching Assistant

- Physical Geology (Lab Instructor), *UArizona* .... Fall 2021, Fall 2023
- Paleontology, *UArizona* .... Spring 2023
- Dinosaurs, *UArizona* .... Fall 2022
- Geochronology and Thermochronology (Guest Lecturer), *MSU* .... Spring 2021
- Introduction to GIS and Cartography (Lab Instructor), *MSU* .... Spring 2020, Spring 2021
- Sedimentation and Stratigraphy (Lab Instructor), *MSU* .... Fall 2019
- Geology Field Camp (Assistant Instructor), *Indiana University/MSU* .... Summer 2019
- Geologic Field Methods, *MSU* .... Fall 2018
- Freshman Seminar, *MSU* .... Fall 2016

Student Mentorship and Training

Mentored as postdoctoral fellow

- William Crater (NSF EAR-PF project; Zircon U-Pb analysis), *ISU, exp. B.S. 2029* 2025–2026
- Tiana Hursh (NSF EAR-PF project; Chile field research), *ISU, exp. B.S. 2027* 2025–2026
- Amarissa Cramer (NSF EAR-PF project; Zircon U-Pb analysis), *ISU, exp. B.S. 2027* 2025–2026
- Parker Hazelbush (NSF EAR-PF project; Chile field research), *ISU, B.S. 2026* 2025–2026

Co-mentored as M.S. and Ph.D. student

- Faisal Sabor (TANGO project; first-author GSA presentation), *UArizona, B.S. 2023* 2023
- Katleho Ramotso (TANGO project; provenance analysis), *UArizona, B.S. 2025* 2023
- Ash Abbate (TANGO project; provenance analysis), *UArizona, exp. B.S. 2026* 2023
- MSU Sedimentary Undergraduate Research program (6 students), *MSU* 2021
- Sophie Black (Montana provenance analysis; coauthor GSA Bulletin), *MSU, B.S. 2021* 2019–2021
- Saré Campbell (Montana provenance analysis; coauthor GSA Bulletin), *MSU, B.S. 2021* 2019–2021
- John Cook (Montana provenance analysis; coauthor GSA Bulletin), *MSU, B.S. 2024* 2019

Field Research (Total fieldwork: > 19 months)

- Coastal Chile Mesozoic forearc basin (6 weeks), *Postdoctoral Research* 2025–2026
- Southern Central Andes, Argentina & Chile (> 9 months), *Ph.D. Research* 2022–2025
- Great Valley forearc basin, CA (6 weeks), *Research Mentor, Research Assistant* 2019–2021
- Southwest Montana (> 7 months), *Field Camp Instructor, M.S. Research* 2018–2021

Service & Outreach

Conference Organization

- Lead Convener, GSA Pardee Symposium, *Bottoms Up: Reconciling bottom-up and top-down approaches to tectonic reconstructions* .... exp. 2026
- Lead Convener, GSA Topical Session, *Pre-Neogene Tectonic evolution of South America* .... exp. 2026
- Instructor, GSA Short Course, *Quantitative analysis, visualization, and modelling of detrital geochronology data* .... exp. 2026
- Co-convener, GSA Topical Session, *Convergent Margin Systems (T160)* 2023
- Assistant Instructor, GSA Short Course, *Detrital Zircon Analytical Methods (SC 512)* 2023
- Organizing Committee, GeoDaze Research Conference, *UArizona* 2022

- Organizing Committee, Seminar Series & Student Colloquium, *MSU* 2019–2021

Peer Review & Editorial Service

- Ad Hoc Reviewer, *US NSF EAR Structure and Physics of the Solid Earth Program* 2026
- Peer Reviewer, *Tectonics, Terra Nova, Geosphere, GSA Bulletin (n = 12 reviews)* 2022–present
- Grant Reviewer, *US–Argentina Fulbright Commission* 2025
- Grant Reviewer, *Graduate and Professional Student Council, UArizona* 2025

University & Departmental Service

- Workshop Presenter, *Ready–to–Rock GSA Prep Series, ISU* 2026
- Panel Member, *Careers and Graduate School in Geoscience, ISU* 2025
- Graduate Student Representative, *Department of Geosciences, UArizona* 2022–2023
- Founding Member, *Decolonize Geosciences, MSU* 2021
- Panel Member, *Tips for Applying to Graduate School, MSU* 2020

Public Outreach & Science Communication

- Contributor, *Science YouTube Channel (@CadenHowlett)* 2022–present
- Science Fair Judge, *Sunrise Drive Elementary School, Tucson, AZ* 2025
- Outreach Presenter, *Legacy Traditional Schools, Tucson, AZ* 2022
- Adaptive Skiing Volunteer, *Eagle Mount Therapeutic Recreation, Bozeman, MT* 2016–2021

Selected Conference Proceedings (23 total since 2019; †Student mentee author)

11. Cramer†, A., W. Crater†, P. Hazelbush†, T. Hursh†, and **C.B. Ronemus** (2026). No Allochthons Allowed: Detrital zircon provenance contradicts Post-Triassic Terrane Accretion in Central Chile. *Idaho State University Research and Creative Works Symposium*. Pocatello, ID.

10. Hursh†, T., P. Hazelbush†, A. Cramer†, W. Crater†, and **C.B. Ronemus** (2026). Basin Development Along the Pre-Andean Margin: Facies architecture of Triassic–Jurassic deposits in Coastal Chile (∼32°S). *Idaho State University Research and Creative Works Symposium*. Pocatello, ID.

9. Howlett, C.J., **C.B. Ronemus**, B. Carrapa, and P.G. DeCelles (2025). Cordilleran orogenic wedge evolution in a transitional segment of the South-Central Andes (34.5°S), Chile and Argentina. *GSA Connects*. San Antonio, TX.

8. Muller, V.A., B. Carrapa, S.N. Thomson, P.G. DeCelles, C.J. Howlett, **C.B. Ronemus**, and S.L. Beck (2025). Exhumation of the Eastern Cordillera, NW Argentina: a record of competing river incision and fold-and-thrust belt propagation. *GSA Connects*. San Antonio, TX.

7. **Ronemus, C.B.**, C.J. Howlett, B. Carrapa, A. Echaurren, M. Barrionuevo, J.G. Mosolf, and M.L. Foley (2025). Crustal thickness evolution of the southern Central Andes (∼35°S): Insights from igneous paleomohometry and zircon petrochronology. *GSA Connects*. San Antonio, TX.

6. **Ronemus, C.B.**, C.J. Howlett, V.A.P. Muller, P.G. DeCelles, B. Carrapa, L. Fennell, J. Suriano, and L. Lothari (2025). Foreland basin evolution in the Andes (∼32°S). *Along Strike Variations in Central Andean Subduction and Mountain Building (NSF Funded Workshop)*. Tucson, AZ.

5. Lothari, L.D., J. Suriano, J.F. Mescua, M. del Llano, A. Echaurren, L.B. Giambiagi, J. Cottle, and **C.B. Ronemus** (2024). El registro sedimentario Paleógeno del retroarco Andino (33–32°S): Extensión hacia el sureste del evento compresivo del Eoceno Temprano. *XXII Congreso Geológico Argentino*. San Luis, Argentina.

4. **Ronemus**, **C.B.**, J. Suriano, C.J. Howlett, and V.A. Muller (2024). La naturaleza de las discontinuidades en la cuenca del antepaís andino: Datos de un nuevo registro sedimentario del Eoceno-Mioceno a $\sim$31.75°S. *XIX Reunión de Tectónica*. San Juan, Argentina.

3. Sabor†, F., **C.B. Ronemus**, and C.J. Howlett (2023). Sedimentologic insights into Late Cretaceous to Early Miocene foreland basin development in the southern Central Andes ($\sim$36°S). *GSA Connects*. Pittsburgh, PA.

2. **Ronemus**, **C.B.** and D.A. Orme (2021). Geologic map of the eastern half of the Melrose 7.5′ quadrangle and the western half of the Wickiup Creek 7.5′ quadrangle, southwestern Montana. *GSA Connects*. (USGS Best Student Map Competition, 1^{st} place). Portland, OR.

1. **Ronemus**, **C.B.**, D.A. Orme, W.R. Guenthner, and S.E. Cox (2021). Orogens of Big Sky country: Reconstructing the deep-time tectonothermal history of the Beartooth Mountains, Montana and Wyoming. *17th Annual Conference on Thermochronology*. Santa Fe, NM.

Industry Experience

- Petroleum Geology Intern, *Matador Resources, Dallas, TX* .... Summer 2021
- Exploration Geologist, *Childs Geoscience Inc., Bozeman, MT* .... part-time 2018–2020
- Mining Geology Intern, *Sibanye-Stillwater, Nye, MT* .... Summer 2018
- GIS Analyst, *Bioresource Consultants, Ojai, CA; MSU Extension, Bozeman, MT* .. part-time 2017–2018

Languages

English (native); Spanish (working proficiency)